Learning Degree-Generalizable Distributed Graph Coloring

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Abstract

Can a learned distributed algorithm operate at graph degrees and palette sizes absent from training? We study synchronous $(\Delta+1)$-coloring using a shared node–color network, Universal Color GNN (UCGNN). Its parameter count is independent of the palette, while its local computation and communication costs grow with it. Historical experiments with five networks trained only on degree-three, 40-node graphs recorded 8,629 successful extreme-study executions, with unequal coverage across degrees. These observations motivate a prospective mechanism study rather than a universal-correctness claim. We isolate prefix normalization at fixed tensor shapes and initialization, compare MAX with SUM and MEAN, and implement prefix-free DeepSets and local palette attention. We distinguish candidate imitation, proposal diversity, authorization safety and progress. A finite-type reduction characterizes the MAX COMMIT channel; a separate anti-concentration condition yields a conditional progress bound. Neither successful rollouts nor local safety alone certify termination of a learned policy. New experimental results are explicitly marked pending until the fixed-budget independent evaluation is complete.

1 Introduction

Classical distributed algorithms specify computations that apply to whole families of graphs. A learned implementation must additionally select a representation whose behavior remains meaningful when neighborhood size changes. Neural algorithmic reasoning investigates learning executable computations and their transfer across inputs [12]; the CLRS benchmark makes this challenge experimentally accessible [13]. Architectural alignment can affect extrapolation, but expressivity and training success do not by themselves establish out-of-distribution correctness [14].

We ask: *Which architectural mechanisms enable learned distributed algorithms to generalize beyond the graph degrees and palette sizes observed during training?* Our case study uses anonymous synchronous nodes, private randomness, two neighbor communications per protocol round, and permanent color decisions. We study a valid coloring with a prescribed palette, rather than optimizing the chromatic number. Local neural computations are simulated on a GPU; we do not claim a network-deployed distributed implementation.

The historical UCGNN result is striking but insufficient to identify causes: five networks trained with four colors produced proper colorings at much larger palettes, while exact candidate imitation degraded. A historical raw prefix variant concentrated proposals and stalled, but also changed the scorer dimension and initialization. Its failure cannot isolate normalization. Similarly, correct ordinary rollouts can hide an unsafe authorization rule when candidate selection rarely exposes conflicts.

Our contribution is an auditable study design and mechanism characterization: (i) a paired intervention on prefix endpoints, (ii) explicit aggregation and palette-processing controls, (iii) independent graph and training-seed evaluation, and (iv) separate conditional arguments for safety

and progress. We make no claim that UCGNN is inherently superior to attention, or that these architectural ingredients are necessary for every learnable coloring policy.

Related work. Message-passing GNNs and distributed local algorithms share locality constraints, although identifiers, message sizes, synchrony and stopping rules matter [4, 7, 10]. Flood and Echo Net aligns propagation with a distributed execution pattern [5]; work on asynchronous GNN inference emphasizes that synchronous performance does not imply asynchronous robustness [9]. Distributed scheduling with learned embeddings followed by a greedy solver is a different interface from learning the entire local decision policy [16]. GNN coloring work has studied decision classification and extraction of constructive solutions [3]; this does not constitute anonymous synchronous protocol execution under our observation constraints. Transformer length generalization depends on the learned computation and input representation [2, 17]. Finally, imitation learning faces policy-induced distribution shift [6]; targeted intervention can expose confounded decision rules [1]. Our counterfactual proposal supervision is specified below and should not be equated with either of these algorithms.

2 Problem and Distributed Model

Let $G = (V, E)$ be a finite simple undirected graph, $N = |V|$, with maximum degree at most a declared Δ. Nodes share the palette $\{0, \dots, K-1\}$, $K = \Delta+1$, and maintain $q_v \in \{-1, 0, \dots, K-1\}$, where -1 is pending. No unique node identifier enters the policy. Colors have common names and an order, so color permutation invariance is not assumed. Each round draws independent $r_v^p, r_v^c \sim U[0, 1)$ even for fixed nodes. After a first neighbor communication the node selects participation B_v and a candidate C_v. A second communication permits a learned COMMIT decision J_v. Participating pending nodes with $J_v = 1$ fix C_v permanently.

The teacher's available set and decisions are

$$A_v = \{c : \forall u \in N(v), q_u \neq c\}, \qquad k_v = |A_v|, \tag{1}$$

$$B_v^* = \mathbf{1}[q_v = -1]\mathbf{1}[r_v^p < 1/2], \qquad C_v^* = A_v[\lfloor r_v^c k_v \rfloor], \tag{2}$$

where A_v is sorted. The teacher confirms only if no fixed neighbor has the candidate and no participating pending neighbor proposes it. We call this the random-trial teacher; historical code sometimes uses "Luby" as shorthand. Exact availability and conflict checks supply labels or observe completed model decisions; they never enter learned inference as a shield.

3 Supervised Learning of Distributed Coloring

Each training example is a round from a teacher trajectory, including the current partial coloring and its two private draws. The loss is

$$\mathcal{L} = \mathcal{L}_{\text{participate}} + \mathcal{L}_{\text{candidate}} + \tfrac{1}{2}\mathcal{L}_{\text{commit,own}} + \tfrac{1}{2}\mathcal{L}_{\text{commit,cf}}.$$

Participation and COMMIT use binary cross-entropy; the candidate uses cross-entropy over the public palette. During ten warmup epochs COMMIT sees teacher proposals. Thereafter it sees hard own proposals and intervened proposals with recomputed exact authorization labels. Interventions include proposing a fixed neighbor's color and equal simultaneous neighboring offers; their frozen probabilities are 0.17 and 0.04. Only proposals are intervened upon: permanent colors and the observation interface remain unchanged. Hard proposals prevent the COMMIT losses from

updating candidate or participation modules. Own-only replaces the counterfactual half by own proposals, preserving the total loss weighting.

M6B uses the earlier M4B neural architecture specialized to a fixed palette. With its reference dimensions it has $385K + 7427$ parameters. Changing K changes input and output matrices, so these checkpoints are evaluated only at compatible degrees, without undocumented padding or adaptation.

4 Universal Color GNN

For every pair (v, c) define $u_v = \mathbf{1}[q_v = -1]$ and $o_{vc} = \mathbf{1}[q_v = c]$. Shared learned MLPs construct the first message and state:

$$m^1_{vc} = \phi_1([u_v, o_{vc}]), \qquad h_{vc} = \psi_1([u_v, \mathrm{MAX}_{w \in N(v)}\, m^1_{wc}]).$$

Both have 32-dimensional nonnegative outputs. Empty-neighborhood aggregation is zero. Learned availability and ordered endpoints are

$$\alpha_{vc} = \sigma(a^\top h_{vc} + b), \qquad S_{vc} = \sum_{j<c} \alpha_{vj}, \qquad D_v = \max(\sum_j \alpha_{vj}, 0.5), \tag{3}$$

$$L_{vc} = S_{vc}/D_v, \qquad U_{vc} = (S_{vc} + \alpha_{vc})/D_v, \tag{4}$$

$$s_{vc} = F([h_{vc}, \alpha_{vc}, L_{vc}, U_{vc}, r^c_v]). \tag{5}$$

The candidate is the maximizing palette slot; there is no exact mask of occupied colors. The availability scores need not be calibrated probabilities or an integer count. Prefix accumulation and normalization are incorporated operations, while availability, scoring and authorization are learned.

For the second communication let

$$f_{vc} = [u_v, o_{vc}, B_v, \mathbf{1}[C_v = c]], \qquad z_{vc} = \psi_2([f_{vc}, \mathrm{MAX}_{w \in N(v)}\, \phi_2(f_{wc})]).$$

The COMMIT head reads only z_{vC_v} and the own participation/pending/offer bits. The original model has 16,035 trainable parameters independent of K. The implemented messages contain $32K$ floats per directed edge per stage; constant parameter count is not constant communication or memory cost.

5 Prospective Experimental Protocol

The protocol was locally committed as `023deaf` before new training or confirmatory test generation. Primary treatments are NORMALIZED, raw endpoints, SUM, MEAN, DeepSets and LocalPaletteTransformer. The raw-endpoint control changes only (L, U) to $(S, S + \alpha)$ in the same four extra scorer inputs. It shares every initial tensor, example, minibatch order, counterfactual tape, optimizer and update count with NORMALIZED within a training seed. SUM and MEAN replace the neighbor reduction in both stages, without extra scaling for SUM; MEAN uses actual in-degree and zero for an isolated node.

Training uses 800 regular graphs with $N = 40$, $\Delta = 3$, three tapes, Adam learning rate 10^{-3}, batch size 128 and 120 epochs. New training seeds are 41000–41004. Seed 41000 is the full-budget pilot and counts as the first replicate. Promotion requires final training-degree validation accuracies of at least 0.98 participation, 0.90 candidate and 0.95 own-COMMIT, with finite training

Model	Networks	Executions	Successes	Graphs	Unsafe
A1	1	400	357	400	0
C1	1	400	400	400	0
E1	5	8629	8629	1765	0
E2	5	2000	2000	400	0
E3	5	2000	2000	400	0

Table 1: Historical extreme-study counts. Coverage differs between experiments; shared graphs do not become independent by evaluating more networks.

throughout. Failed pilots are retained without repeated tuning. Promotion decisions precede the confirmatory test. Epoch 120 is primary; validation-best is a separately labeled selection policy.

The independent namespace 202610210211 fixes 304 graph–tape cases per model: degrees 3,5,10,15,20,50,100,200, regular/irregular/heterogeneous families, $N = 160$ (400 at degree 200), size checks at $N = 640$, cliques and circulant bipartite stress graphs. Each random core cell contains eight graphs; size cells contain four. A clique's three tapes are not three independent graphs. Every paired model uses the same instances and tapes, with a 256-round cap. Failures do not stop other evaluations or truncate a rollout at its first unsafe action. Technical OOM and timeout remain distinct from wrong coloring.

We report fixed-network Clopper–Pearson intervals per random-graph stratum, all seed rates, and paired hierarchical bootstrap differences across training seeds and graph identities. Correlated executions on shared graphs are not pooled as independent Bernoulli trials. Computational limits are an 8 GiB RTX 4060, one GPU worker, a 4 GiB allocator cap, and 36 accumulated active GPU-process hours. Source hashes, optimizer/RNG checkpoints and atomic per-case records support recovery. The full preregistration specifies the remaining metrics and resource rules.

6 Degree and Size Generalization: Historical Evidence

Five historical E1 networks trained solely at degree three recorded 8,629/8,629 successes in the extreme study, on 1,765 distinct graphs with unequal coverage across degree, size and seed. The maximum degree 500 was exploratory with sparse coverage, not a balanced five-network test at that degree. Historical E2 and E3 each had five independent training networks; A1 and C1 each had one. These already observed cohorts are motivation and historical evidence, not untouched architecture-selection tests.

E1-0's exact candidate fidelity can collapse while valid coloring remains successful. In a historical all-free $K = 201$ controlled sweep it agreed with the teacher at only 5/1,001 uniform-grid draws while using 199 colors. Thus candidate identity and diversity are different endpoints. Historical A1's all-free $K = 101$ law placed mass on only four colors. Reconstructed failures showed extremely slow progress with no unsafe commits, localizing a functional failure without isolating which architectural change caused it.

TODO: Awaiting independent confirmatory degree/size results for the new models.

7 Architectural Ablations

TODO: Awaiting confirmatory architectural ablation results.

The isolated raw control addresses normalization under one fixed training procedure. A successful raw control would refute necessity under that procedure; a negative result would not establish

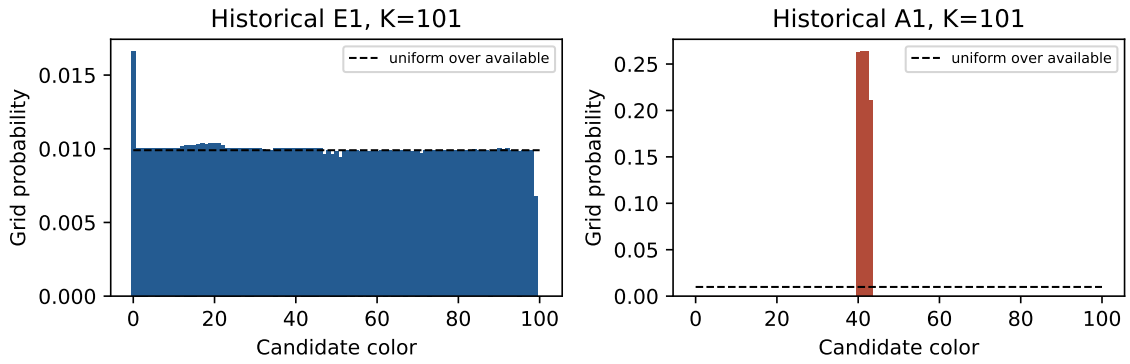


Figure 1: Historical candidate distributions, generated from saved per-color CSVs. These descriptive observations do not isolate the effect of normalization.

Model	Seed	Epoch	Params	Updates	Candidate	COMMIT
deepsets	41000	120	17907	17880	0.9948	1.0000
mean	41000	120	16035	17880	0.9983	1.0000
normalized	41000	120	16035	17880	0.9971	1.0000
normalized	41001	120	16035	17640	0.9985	1.0000
normalized	41002	120	16035	17760	0.9986	1.0000
normalized	41003	120	16035	17760	0.9985	1.0000
normalized	41004	120	16035	17760	0.9993	1.0000
own_only	41000	120	16035	17880	0.9971	1.0000
raw_endpoints	41000	120	16035	17880	0.9850	1.0000
raw_endpoints	41001	120	16035	17640	0.9897	1.0000
raw_endpoints	41002	88	16035	13024	0.9852	1.0000
sum	41000	120	16035	17880	0.9983	1.0000
transformer	41000	120	16995	17880	0.9957	1.0000

Table 2: Training progress and in-degree validation, with all seeds shown. Partial epochs are explicitly listed.

impossibility with all other parameterizations or training budgets. MAX removes multiplicity but retains which message types occur. SUM retains counts and changes magnitude; MEAN retains type proportions. No outcome is assumed in advance.

8 Transformer and Other Baselines

LocalPaletteTransformer keeps shared local graph communication and the COMMIT pathway, replacing the prefix scorer by a 16-dimensional, two-head within-node self-attention block with feedforward width 32 and zero dropout [11]. It receives functional positions $c/\max(K-1,1)$ and $1/K$, not learned absolute position embeddings or original prefixes. Its parameter count is 16,995. The DeepSets control uses a shared 16-dimensional mean-pooled palette context [15], functional positions and 17,907 parameters. Attention can have quadratic palette cost; a technical resource limit must not be called a coloring failure. All six seed-41000 pilots reached epoch 120 and passed the prespecified local learning gate. Candidate validation fidelity was 0.9971 for NORMALIZED, 0.9850 for raw endpoints, 0.9983 for SUM and MEAN, 0.9948 for DeepSets and 0.9957 for the Transformer;

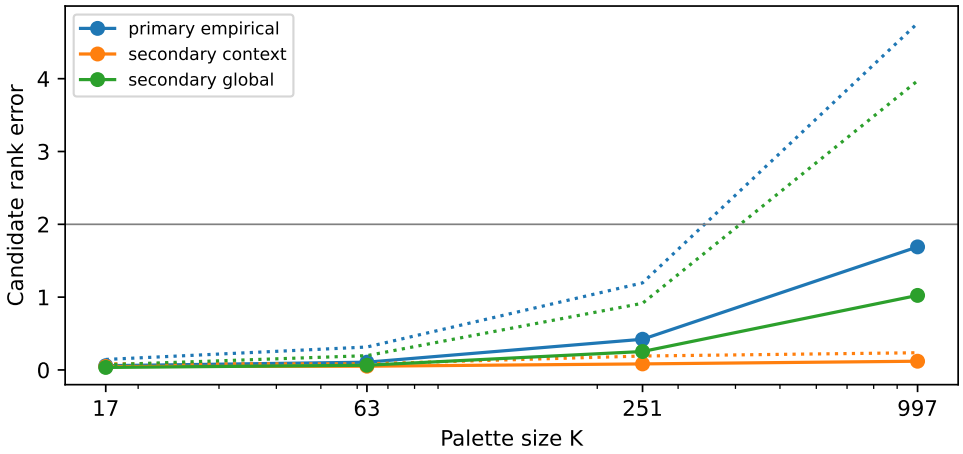


Figure 2: Frozen E1-0 candidate predictions from saved CSVs. Solid lines show mean per-state rank error; dotted lines show the worst per-state mean. Primary and secondary predictors remain separately identified.

own-proposal COMMIT fidelity was 1.0 for each. These are in-degree validation measurements, not degree-transfer results. In a separate inference-only resource check, all six models fit at $N = 640$, $\Delta = 200$ and $K = 201$. The Transformer used approximately 532 MiB of peak allocated CUDA memory for one round; this establishes technical feasibility for this configuration only.
TODO: Awaiting Transformer/DeepSets degree-transfer and candidate-law results.

These are architectural controls, not a claim that the only possible prefix-free model has been tested. Without positions or another symmetry breaker, identical available slots receive identical scores under an invariant pooled context, preventing general ordered-teacher imitation.

9 Understanding Learned Policies

9.1 Selection and held-out prediction

The prospective diagnostic freezes a monotone rank calibration of historical E1-0 on all-free $K = 101$ before evaluating $K = 17, 63, 251, 997$ and low, high, scattered, alternating and extreme availability patterns. It reports rank errors, forbidden proposals, total variation from uniform, entropy, effective colors, maximum mass and pair collisions. Boundary-neighbor float32 probes are separate from the midpoint grid used for distribution estimates. The CPU float32 diagnostic, prospectively authorized in addendum `8c69168`, evaluated 120 configured states with 2,049 midpoint draws each. Equivalent all-free configurations occur under multiple pattern names; these are not 120 independent random states. No forbidden candidate was observed on these grids. The frozen empirical predictor's mean per-state rank error was 0.569; 10 configurations at $K = 997$ exceeded two, with a worst per-state mean of 4.761. Thus its approximation deteriorates at the largest palette despite a small pooled mean. We retain these negative cases.

Two secondary analytic predictors were also frozen before held-out inspection. The context-conditioned predictor uses a finite catalog of neighbor-type contexts calibrated at $K = 101$. It achieved mean rank error 0.075 and worst per-state mean 0.236, with individual grid rank error at most one. This is finite empirical evidence for a context-dependent deformed quantile, not an exact algorithmic identity or a guarantee over all availability patterns.

9.2 A sufficient condition for progress

Condition on a proper state. Let U_v be the pending neighbors. With private independent participation/candidate draws within and across nodes, and exact conflict rejection,

$$\Pr(v \text{ confirms} \mid \text{state}) = p_v \sum_{c \in A_v} q_v(c) \prod_{u \in U_v} (1 - p_u q_u(c)). \tag{6}$$

Confirmation events of neighboring nodes need not be independent.

Proposition 1. *Suppose $0 < p_{\min} \le p_u \le p_{\max} < 1$, exact conflict rejection and the independence assumptions of (6) hold. If $\operatorname{supp}(q_v) \subseteq A_v$ and $\max_c q_v(c) \le \beta/k_v$, then*

$$\Pr(v \textit{ confirms} \mid \textit{state}) \ge p_v(1 - p_{\max})^{\beta |U_v|/k_v} \ge p_{\min}(1 - p_{\max})^{\beta}.$$

Proof. Properness and $K \ge \Delta + 1$ imply $k_v \ge |U_v| + 1$. For $0 \le x \le p_{\max}$, concavity gives $\log(1 - x) \ge x \log(1 - p_{\max})/p_{\max}$. Set $\Lambda_c = \sum_{u \in U_v} p_u q_u(c)$ and $a = \log(1 - p_{\max})/p_{\max} < 0$. The product in (6) is at least $\exp(a\Lambda_c)$. Jensen's inequality gives $\sum_c q_v(c) e^{a\Lambda_c} \ge e^{a \sum_c q_v(c)\Lambda_c}$. Moreover, $\sum_c q_v(c)\Lambda_c \le (\beta/k_v) \sum_u p_u \le \beta p_{\max} |U_v|/k_v$. Since $a < 0$, substitution proves the first bound; $|U_v|/k_v \le 1$ proves the second. □

If these hypotheses hold uniformly at every reachable state for every pending node, $\rho = p_{\min}(1 - p_{\max})^{\beta}$ gives $\Pr(T > t) \le \min\{1, N(1-\rho)^t\}$ by conditional iteration and union bound. We have not established the uniform candidate hypotheses for any learned checkpoint. In contrast, proposing one common deterministic color on an all-free K-clique with independent participation $1/2$ is safe but has mean time $2^K/K$ until the first confirmation. Uniform *marginals* with a shared candidate draw have the same failure of progress, showing why conditional independence matters.

9.3 Finite-type safety of COMMIT

Proposition 2. *For a legal uncolored participating center with candidate c, nonnegative shared messages and MAX aggregation, the UCGNN COMMIT logit depends only on the set of six legal neighbor types in slot c. Correct strict signs on all 64 type sets imply exact local conflict rejection for arbitrary degree and palette in real arithmetic.*

Proof. The legal types $[u, o, B, \text{offer}]$ are

$$(0,0,0,0), (0,1,0,0), (1,0,0,0), (1,0,0,1), (1,0,1,0), (1,0,1,1).$$

Colored neighbors cannot participate or offer; pending neighbors do not own a fixed color. MAX is idempotent, so multiplicities disappear. The own candidate gather reads one slot, whose own features are $(1,0,1,1)$; other colors cannot alter that logit. Fixed and simultaneous conflicts correspond precisely to the second and sixth types. The 64 strict sign conditions therefore suffice. Other own states are prevented from execution by explicit protocol gating. All subsets are realizable by proper star states with at least two colors. □

Induction then preserves properness, without proving progress or termination. All five historical E1 checkpoints now have interval-certified strict signs on all 64 sets, both for real weights and under the stated float32 affine-error model. Their real-arithmetic minimum signed margins range from 9.706 to 13.547. Historical C1 and the new own-only seed-41000 checkpoint each have 16 certified unsafe type sets. The new NORMALIZED, raw-endpoint, DeepSets and Transformer seed-41000 COMMIT channels also satisfy the certificate; this does not certify their candidate selection. SUM

and MEAN lack this 64-set reduction, which is an applicability limitation rather than evidence of unsafe decisions. Complete arithmetic assumptions, per-mask intervals and checkpoint hashes accompany the certificate CSV and JSON records.
TODO: Awaiting remaining checkpoint certificates, numeric perturbation and controlled rollout progress probes.

10 Discussion, Limitations and Future Work

Success on finite graphs is evidence about a specified distribution, rather than universal correctness. Five training networks permit only limited inference over initialization; small graph strata have wide confidence intervals. Shared graphs and repeated clique tapes must retain their dependence in analysis. The generic baselines retain UCGNN's graph-interface and safety pathway, so they do not test every possible end-to-end attention architecture. A fixed parameter count still permits costs that grow with N, $|E|$ and K.

An exploratory alternative teacher starts at $\lfloor Kr_c \rfloor$ and scans cyclically to the first available color, keeping participation and rejection unchanged. Its candidate mass at c is the cyclic gap preceding c divided by K, usually different from uniform over available colors.
TODO: Awaiting fresh-from-random-initialization alternative-teacher pilot.

Teacher-free RL and supervised-to-RL fine-tuning ask separate questions; the accompanying design specifies rewards, private stochastic actions and a PPO-based comparison [8], with no large RL run here. Asynchronous execution, compressed communication, certified candidate anti-concentration and larger independent cohorts remain future work.

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